Shri Vile Parle Kelavani Mandal's
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING
(Autonomous College Affiliated to the University of Mumbai)
NAAC Accredited with "A" Grade (CGPA : 3.18)

JAVA ASSIGNMENT

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BRANCH:AIML

BATCH:A1

DATE OF SUBMISSION:28/03/24

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AI-ML

JAVA :- Exp - 2

Aim/objective:-
Implement Arrays in JAVA

Description of Experiment

- Explain Arrays in Java

→ An array is an object which contains elements of a similar data type. Additionally, the elements of an array are stored in a contiguous memory location. It is a data structure where we store similar elements. We can store only a fixed set of elements in JAVA array. It is index-based, the first element on 0th index, 2nd element is stored on 1st index and so on.

- Difference between arrays in Java and C.

C

1) Syntx of array: datatype arrayName [array size]
2) Size of array is typically fixed and cannot be changed its declared.
3) No-inbuit boundary checking of array.
4) Multidimension Arrays are implemented as array of array.
5) When passing an array to function, a pointer to the first element of the array is passed

JAVA

1) syntax of array:- datatype arrayname [] = new datatype [];
2) arrays are dynamically sized objects and their size can be

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AL runtime using classes such as 'ArrayList'.

3] Array bound checking is performed by the JVM at runtime and an 'ArrayIndexOutOfBoundsException' will be thrown if you to access an index outside the valid range.

4] Multidimensional array are arrays of higher dimensional array.

5] Array are object and when passed to methods, a reference to the array is passed.

- Syntax for declaring 1D, 2D array in java –

1-D array (n-length)

Syntax: datatype arrayname[] = new datatype[n] or
datatype [] arrayname[] = new datatype[n]

2-D array (n,m → length or size)

~~dat~~ data type arrayname[][] = new datatype[n][m] or
data type [][] arrayname = new datatype[n][m]

- Explain the concept of irregular arrays :–

Irregular array in java is also known as Jagged array. It is a collection of arrays where each array may contain a varied number of elements. A two-dimensional array, in contrast, require all rows and column to have same length. They can be created by specifying the size of each array in declaration, for eg: –a jagged array with three rows can have the first row with three elements, second row with 2 elements and 3rd with 4 elements.

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- Array length instance variable of array :-

→ Array have a public instance variable called length, which gives you the length of the array. Generally you should avoid using public instance variable to improve encapsulation. However, Java Arrays are thing so we must use them as well.

for eg: -

```
int [] arr = {1, 3, 5, 7, 9}
for (int i = 0; i < arr.length; i++)
    {
        System.out.println(arr[i]);
    }
```

above eg :- Length of arr is 5.

the output is :-

1
3
5
7
9

Conclusion :- In this experiment, I learnt how to implement arrays in Java.

- To implement Arrays :-

You have been given arrays of positive integers $A_1, A_2, \ldots A_n$ with length n and you have to point an array

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of same length (N), where the values in the new array are the sum of every number in the array except the number at that index

Program: —

```
import java.util.*;
class Array
{
 public static void main (String args[]){
 int n, i, sum=0;
 System.out.print("Enter the number of elements");
 Scanner s=new Scanner(System.in);
 n = s.nextInt();
 int a[] = new int[n];
 int b[] = new int[n];
 for (i=0; i<n; i++)
 {
 System.out.print("enter the value of a[i]: ");
 a[i] = s.nextInt();
 }
 for (i=0; i<n; i++)
 {
  sum=sum + a[i];
 }
 for (int i=0; i<n; i++)
 {
  b[i]=sum-a[i]; }
```

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5

```
for(i=0; i<n; i++)
{
System.out.println(" The sum of value of b[i]
+ b[i]);
}
}
```

Output:-

```
Enter the name of elements 5
Enter the value a[i] :1
enter the value a[i]:2
enter the value a[i]:3
enter the value a[i]=4
enter the value a[i]=5
the sum of value at b[i] 14
the sum of value at b[i] 13
the sum of value at b[i] 12
the sum of value at b[i] 11
the sum of value at b[i] 10
```

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Experiment No: 2

a. You have been given an array of positive integers A1, A2,...,An with length N and you have to print an array of same length (N) where the values in the new array are the sum of every number in the array, except the number at that index.

i/p 1 2 3 4

For the 0th index, the result will be 2+3+4= 9, similarly for the second, third and fourth index the corresponding results will be 8, 7 and 6 respectively.

i/p 4 5 6

o/p 11 10 9

```
import java.lang.*;
import java.util.Scanner;


class SumOfArrayelements
{

        public static void main(String args[])
        {
                int number,a[],b[];
                Scanner sc=new Scanner(System.in);
                System.out.println("Enter Number of Elements(SIZE of Array) in array: ");
                number=sc.nextInt();

                //Construct an array of above size
                a=new int[number];
                b=new int[number];
                //Input Elements in Array
```


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```
System.out.println("Enter "+number+" Elements in Array:");
for(int i=0;i<a.length;i++)
        {
                System.out.print("Enter Element at a["+i+"]: ");
                 a[i]=sc.nextInt();
        }

//logic to find sum of all elements in array except the index element
int sum=0;
for(int i=0;i<a.length;i++)
{
        sum=0;
        for(int j=0;j<a.length;j++)
        {
                if(i!=j) //condition to exclude the index element from sum
                        sum=sum+a[j];
        }
        b[i]=sum;


}
System.out.print("Elements from Input Array:");
for(int i=0;i<a.length;i++)
        {
                 System.out.print(a[i]+" ");
        }
System.out.println();
System.out.print("Elements from Output Array:");
for(int i=0;i<a.length;i++)
        {
                 System.out.print(b[i]+" ");
        }

    }

}
```


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/*
OUTPUT:

```
C:\kavya\java\College Pratice Q\Exp 2 Code>javac SumOfArrayelements.java

C:\kavya\java\College Pratice Q\Exp 2 Code>java SumOfArrayelements
Enter Number of Elements(SIZE of Array) in array:
5
Enter 5 Elements in Array:
Enter Element at a[0]: 1
Enter Element at a[1]: 2
Enter Element at a[2]: 3
Enter Element at a[3]: 4
Enter Element at a[4]: 5
Elements from Input Array:1 2 3 4 5
Elements from Output Array:14 13 12 11 10
C:\kavya\java\College Pratice Q\Exp 2 Code>
```


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b.The annual examination results of 5 students are tabulated as follows:

Roll No	Subject1	Subject2	Subject3

WAP to read the data and determine the following
Total marks obtained by each student
The student who obtained the highest total marks */

```
import java.lang.*;
import java.util.Scanner;
class AnnualExamResult
{

        public static void main(String args[])
        {
                int number,std[][],max[],total=0,j;
                Scanner sc=new Scanner(System.in);
                System.out.println("Enter Number of Students: ");
                number=sc.nextInt();

                //Construct an array of above size
                std=new int[number][5];
                max=new int[5];
                //Input Elements in Array
                System.out.println("Enter "+number+" Student Exam record:");

                for(int i=0;i<std.length;i++)
                        {
                                total=0;
                                System.out.print("Enter Studnet "+(i+1)+" Roll No & Marks of 3
Subjects: ");
```


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```
                    for(j=0;j<4;j++)

                    {

                        std[i][j]=sc.nextInt();
                        if(j!=0)
                            total=total+std[i][j];
                    }
                    std[i][j]=total;
                    max[i]=total;

                }
            System.out.println();
            //Student Record
            System.out.println("RollNo \tSubject1\tSubject2\tSubject3\t TotalMarks");
            for(int i=0;i<std.length;i++)
                {
                    for(j=0;j<5;j++)
                    {
                        System.out.print(std[i][j]+"\t\t");

                    }
                    System.out.println();

                }
            //Logic to find Maximum marks obtained by the student
            int maxMarks=0,r=0;
            for(int i=0;i<max.length;i++)
                {
                    if(max[i]>maxMarks)
                    {
                        maxMarks=max[i];
                        r=i; //recording roll number of studnet who has got
maximum marks
                    }

                }
            System.out.println("Record of studnet who have Maximum Marks: ");
            System.out.println("RollNo \tSubject1\tSubject2\tSubject3\t
            TotalMarks");
```


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```
            for(j=0;j<5;j++)
                {
                        System.out.print(std[r][j]+"\t\t");

                }


        }

}
```

```
Enter Number of Students:
5
Enter 5 Student Exam record:
Enter Student 1 Roll No & Marks of 3 Subjects: 1 22 2 3
Enter Student 2 Roll No & Marks of 3 Subjects: 2 43 23 65
Enter Student 3 Roll No & Marks of 3 Subjects: 3 73 43 37
Enter Student 4 Roll No & Marks of 3 Subjects: 4 12 23 45
Enter Student 5 Roll No & Marks of 3 Subjects: 5 23 34 45

RollNo  Subject1        Subject2        Subject3        TotalMarks
1               22              2               3               27
2               43              23              65              131
3               73              43              37              153
4               12              23              45              80
5               23              34              45              102
Record of student who have Maximum Marks:
RollNo  Subject1        Subject2        Subject3        TotalMarks
3               73              43              37              153
PS C:\Users\thehe>
```


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c. WAP to display following pattern using irregular arrays (jagged arrays). 1
1 2
1 2 3 ……..

```
import java.util.*;
public class Pattern
{
 public static void main(String[] args)
 {

        int n;
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter Number of Rows in Pattern: ");
        n=sc.nextInt();

        int[][] p = new int[n][];

        for(int i=0;i<n;i++)
                p[i] = new int[i+1];

 //initializing array
 for(int i=0; i < p.length; i++)
        {
        for(int j=0; j < p[i].length; j++)
                {
 p[i][j] = j+1;
                }
 }


 //printing Pattern Using Jagged Array
 for(int i=0; i < p.length; i++)
        {
        for(int j=0; j < p[i].length; j++)
                {
 System.out.print(p[i][j]+" ");
 }
                System.out.println();
```


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```
        }

  }
}
```

OUTPUT:

```
C:\kavya\java\College Pratice Q\Exp 2 Code>javac Pattern.java

C:\kavya\java\College Pratice Q\Exp 2 Code>java Pattern
Enter Number of Rows in Pattern:
4
1
1 2
1 2 3
1 2 3 4

C:\kavya\java\College Pratice Q\Exp 2 Code>
```


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